Soft Condensed Matter: An Introduction (2025)

Lecture 1: Different Faces of Soft Condensed Matter

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Web version

Slides

PDF *32 MB* · PowerPoint *146 MB* · Keynote *160 MB*

An introduction to soft materials, why they're soft, and how we measure softness. See course overview and structure for context.

Prologue: Opening Poll

"Name one soft material you've touched today."

Soft matter is everywhere in daily life—from the water in your shower and shampoo to the bread on your table, tissues, foams, and gels. This simple question anchors a deeper truth: the materials we interact with most are soft.

What is Soft Matter?

Definition

Materials easily deformed by modest stresses, thermal fluctuations, or interfacial forces; typically organized by mesoscopic building blocks (droplets, bubbles, grains, polymer coils) held together by relatively weak interactions.

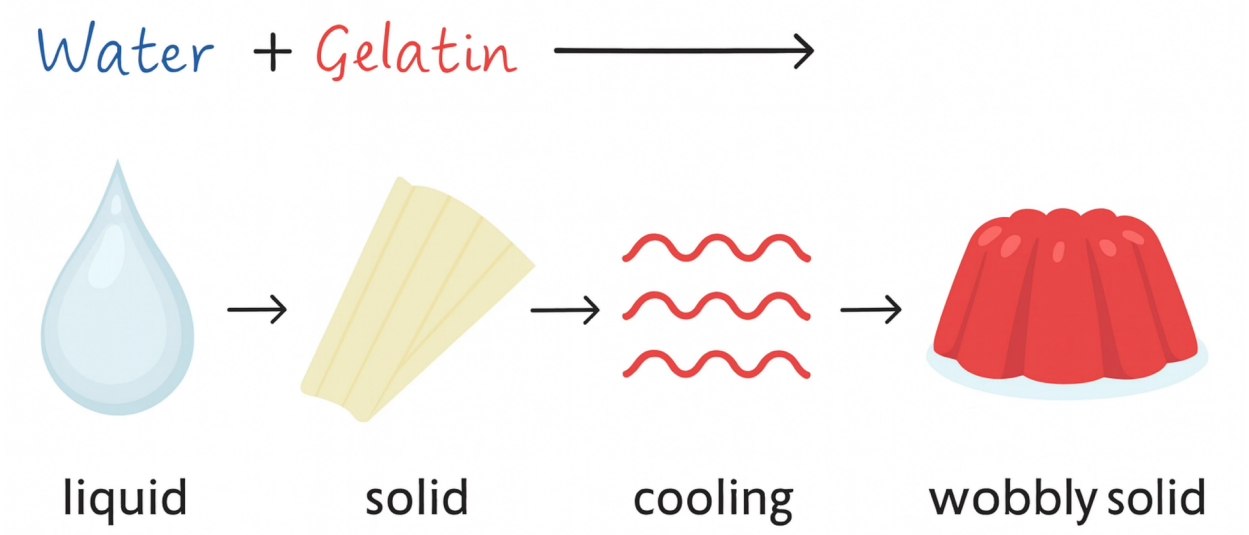
Everyday Snapshots

Water + Gelatin → Gel

Observe the transformation of gelatin and water:

- Liquid state: Individual molecules move freely
- Cooling: Gelatin chains form a crosslinked network
- Solid state: Wobbly, flexible gel structure
- Mechanism: Structure held by *weak* (often reversible) interactions, not rigid covalent bonds

The gel feels solid (can hold shape), yet wobbles easily when pushed—a hallmark of soft matter.



Beer Foam (Foam)



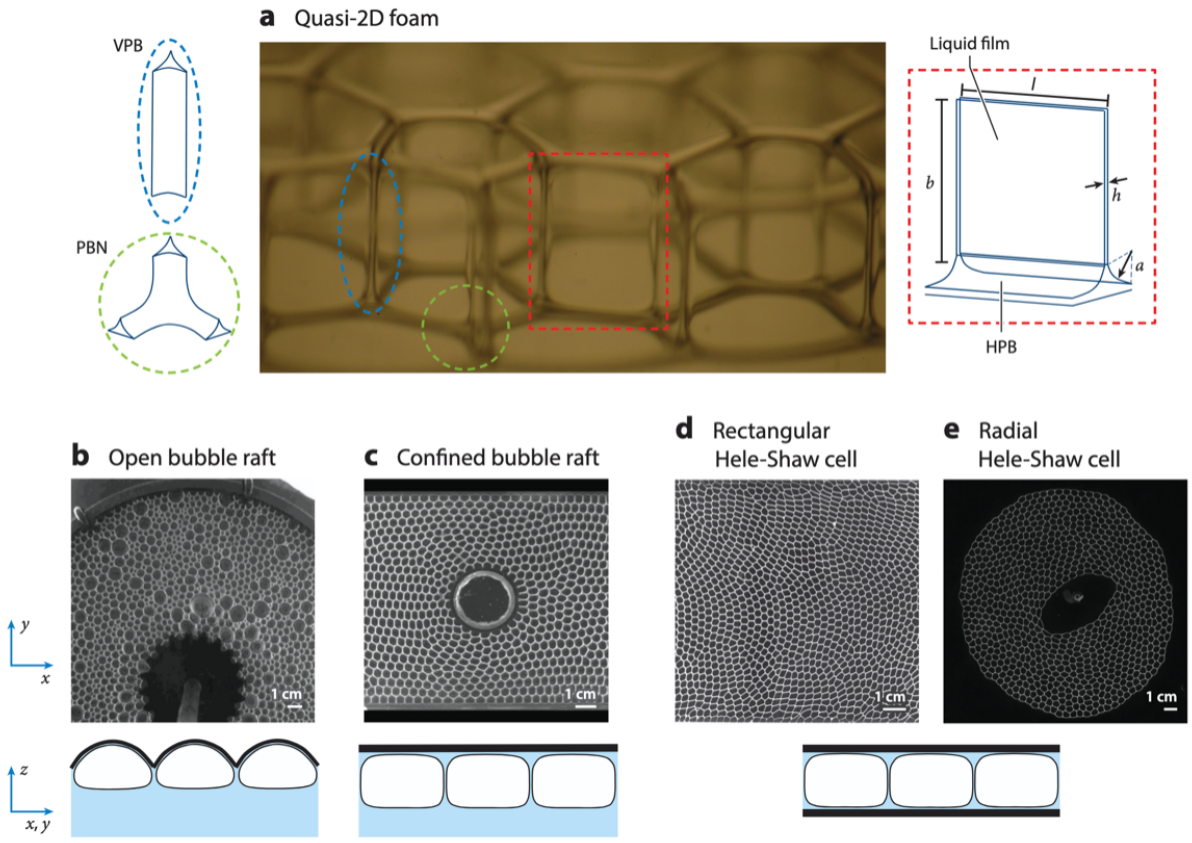


Good foam (left): Stable bubble structure with clear thin walls

Bad foam (right): Bubbles coalesce and collapse

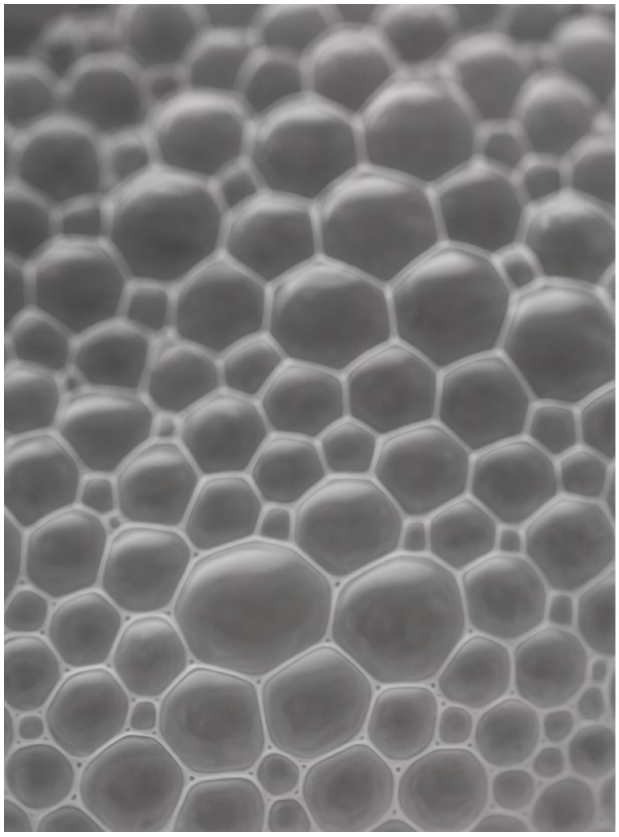
Foams are gas bubbles trapped in liquid with surfactants stabilizing the interfaces. They exemplify soft matter because:

- Structure held by surface tension (weak force)
- Easy to deform under modest stresses
- Flow and rearrange under sustained forces
- Evolve over time via coalescence and coarsening



Reference: P. S. Stewart & S. Hilgenfeldt, *Annu. Rev. Fluid Mech.*, 55:1, 323–350 (2023)

Shaving Cream





A dense foam of gas + liquid—microstructure determines function:

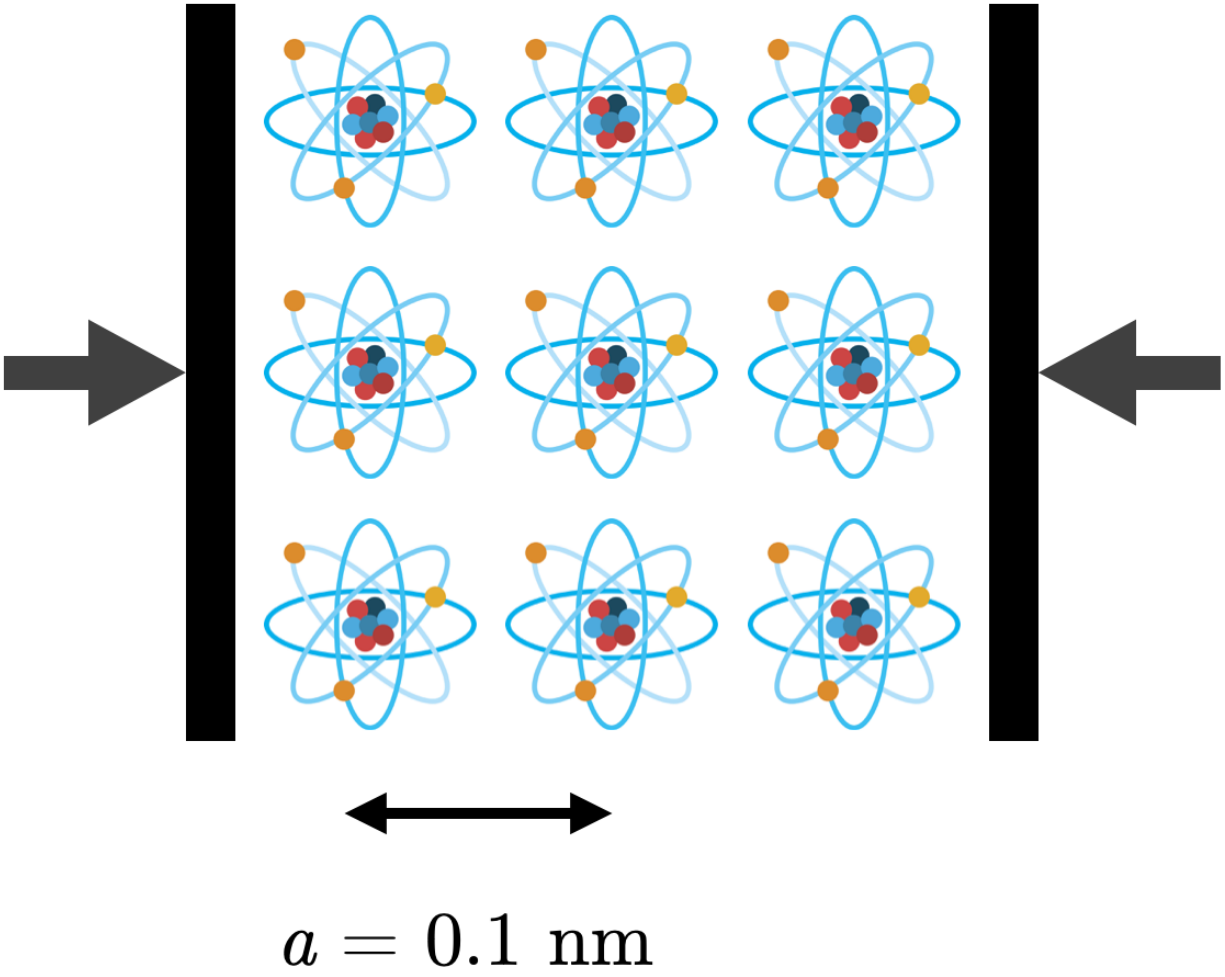
- Spreads easily (flows under shear)
- Holds shape temporarily (stores energy)
- Responds to manual deformation (viscoelastic behavior)

Key Insight

In many soft materials, surface tension, entropy, and geometry govern structure and mechanics far more than stiff, covalent bond stretching does.

Why is "Hard Matter" Hard?

Atomic Scaling Argument



When you compress a solid, you push overlapping electron clouds closer together. The bulk modulus scales roughly as:

$$K \sim \frac{U}{a^3} \tag{1}$$

where:

- U = interaction energy per atom (~tens of eV, from Coulombic/quantum mechanics)
- a = atomic length scale (~1 Ångström)

This is the correct scaling: interaction energies packed into atomic volumes yield enormous bulk moduli.

Numerical Estimate

Coulombic energy at 1 Ångström:

Two unit charges separated by $r = 1\,\text{Å}$ give:

$$U = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{1\,\text{Å}} \approx 14.4\,\text{eV} \tag{2}$$

Converting to modulus:

Using the conversion $1\,\text{eV/Å}^3 = 160.22\,\text{GPa}$:

$$K \sim \frac{14.4\,\text{eV}}{(1\,\text{Å})^3} \approx 2.3 \times 10^6\,\text{MPa} \tag{3}$$

(*Higher estimates using Bohr radius $a_0 \approx 0.529\,Å$ and $U \approx 27.2\,eV$ reach $\sim 2.9 \times 10^7$ MPa—the key point is the scaling, not the exact prefactor.*)

Reality Check: Bulk Moduli of Real Materials

Material	K (MPa)	K (GPa)	Note
Diamond	4.4×10^5	440	Hard to compress and hard to shear
Steel	$(1.6\text{–}1.7) \times 10^5$	160–170	Hard to compress and hard to shear
Water	2.2×10^3	2.2	Hard to compress, liquid (no static shear)
Rubber	$(1\text{–}2) \times 10^3$	1–2	Hard to compress, soft in shear

Important

Most condensed matter is hard to compress (large K). What makes something feel "hard" in everyday life is actually the shear modulus G, not the bulk modulus. Everyday "hardness" ≈ resistance to shape change, not resistance to compression.

Why is Soft Matter *Soft*? — The Entropic Shear Modulus

For an ideal network of flexible polymer strands (rubbers and many swollen polymer gels), the small-strain shear modulus arises from entropy, not from stiff bond stretching:

$$\boxed{G = nk_BT \quad \text{or} \quad G \sim \frac{k_BT}{\xi^3}} \tag{4}$$

where:

- n = number of elastically active strands per unit volume (strands joined to the network at both ends; dangling ends and loops do not count)
- ξ = mesh size of the network, so that $n \sim \xi^{-3}$ (about 1–2 nm in rubbers, 10–100 nm in soft gels)
- k_BT = thermal energy (~4.1×10^{-21} J at room temperature)

Note

$G = nk_BT$ is the ideal-network result; real networks differ from it by factors of order one. The form $G \sim k_BT/\xi^3$ is its scaling version for a network of mesh size ξ. An entangled semi-dilute polymer solution follows the same scaling with concentration, but its plateau modulus is about a hundred times smaller, because the spacing between entanglements is about ten times the blob size ξ; see Polymer-Concentration-Regimes-and-Blob-Networks for details.

Not all soft matter is soft for this reason. In foams and concentrated emulsions, the shear modulus comes from surface tension: shearing the packing of bubbles or drops stretches their interfaces, so $G \sim \gamma/R$, where γ is the surface tension and R the bubble or drop radius, up to a prefactor smaller than one that depends on the liquid fraction. For $\gamma \approx 30\,\mathrm{mN/m}$ and $R \approx 30\,\mu\mathrm{m}$, $\gamma/R \approx 1\,\mathrm{kPa}$: enough for shaving cream to hold its shape, with no entropic elasticity involved.

Microscopic Origin: Stretching Reduces Configurations

At the chain level:

- Stretching a polymer coil reduces the number of configurations it can adopt
- Entropy decreases when stretched: $\partial S/\partial x < 0$ for an extension $x > 0$
- For an ideal chain, the internal energy U does not depend on x, so at fixed temperature the free energy $F = U - TS$ changes only through the entropy. The force the chain exerts on whatever holds its ends, the restoring force, is

$$f_{\text{chain}} = -\frac{\partial F}{\partial x} = T\frac{\partial S}{\partial x} < 0, \tag{5}$$

  which pulls the ends back together. Holding the chain at extension x requires the opposite external force, $f_{\text{ext}} = -T\,\partial S/\partial x > 0$
- Unlike a metal spring (stiff atomic bonds resist), this force is statistical

From one chain to a network:

Crosslink the chains into a network with n elastically active strands per unit volume ⇒ the elastic free-energy density is:

$$F/V \sim nk_BT \quad \Rightarrow \quad G = nk_BT \tag{6}$$

For nearly incompressible materials (Poisson's ratio $\nu \to 0.5$):

$$E \approx 3G \tag{7}$$

Practical Examples

With $k_BT \approx 4.1 \times 10^{-21}\,\mathrm{J}$ at room temperature:

$$G \sim \frac{k_BT}{\xi^3} \quad \Rightarrow \quad \begin{cases} \xi = 10\,\mathrm{nm} & \Rightarrow G \approx 4\,\mathrm{kPa} \\ \xi = 1\,\mathrm{nm} & \Rightarrow G \approx 4\,\mathrm{MPa} \end{cases} \tag{8}$$

Interpretation: Small changes in mesh size (network crosslink density) move the shear modulus from the kPa range (very soft, like a loose gel) to the MPa range (stiffer, more rubber-like), while the bulk modulus stays large (still hard to compress).

The Soft Matter Paradox Resolved

Soft in shear, hard in bulk. Stretching a polymer network just rearranges coil conformations (easy, entropy-driven). Compressing it still pushes on dense electron clouds and liquid trapped in pores (hard, bonding physics). This is why rubbers and gels are "soft" yet "incompressible".
Two caveats. A gel is incompressible only on times too short for its solvent to flow through the network; given time, the solvent drains and the network alone can change volume. Foams are compressible: their gas sets a bulk modulus of order the gas pressure, about 0.1 MPa, far below the GPa values of liquids and rubbers.

Far from Equilibrium: Dynamics Dominate

Soft matter reveals its richness when driven away from equilibrium. Time and forcing history often matter more than equilibrium structure alone.

Drop Impact in Microgravity (ISS "Liquid Ping-Pong")

Video: `https://www.youtube.com/watch?v=TLbhrMCM4_0`

Without gravity, a water droplet maintains a sphere. When two droplets collide:

- Surface tension dominates (gravity suppressed)
- Complex coalescence dynamics unfold
- Demonstrates soft matter far from equilibrium

Mud Volcano at Yellowstone

Video: `https://www.youtube.com/watch?v=a9hUsVq9q7U`

Mud is a elastoviscoplastic soft material:

- Behaves as solid over short times (can support surface bumps and cones)
- Flows like liquid over long times (material spreads slowly)
- Time-dependent response is essential to understanding behavior

Champagne Bubbles: A Glass of Soft Matter

Video: `https://www.youtube.com/watch?v=7HNv5cLObTY`

Bubbles rising through champagne display remarkable dynamics driven by:

- Liquid-gas interfaces (surface tension γ)
- Dissolved CO_2 (pressure inside bubbles)
- Surfactants from wine components
- Viscoelasticity of the liquid (if polymers are present)

Key observation: Adding a small amount of long-chain polymer changes what happens after the bubble bursts, not the bubble itself. In the high-speed movies shown in the lecture (experiments by Alexandros T. Oratis, Physics of Fluids, University of Twente), the bubbles look the same before bursting. After the cap ruptures, the Worthington jet becomes shorter as the polymer concentration increases, drops stay tethered to the jet by thin polymer threads, and at the highest concentration no jet forms at all. The polymer chains resist the strong stretching flow in the jet. Simulations of bubble bursting in viscoelastic liquids show the same suppression of jets and droplets as the elasticity of the liquid increases.

References:

- E. Ghabache, G. Liger-Belair, A. Antkowiak & T. Séon, *Sci. Rep.*, 6, 25148 (2016)
- E. Ghabache, A. Antkowiak, C. Josserand & T. Séon, *Phys. Fluids*, 26(12) (2014)
- P. L. L. Walls, L. Henaux & J. C. Bird, *Phys. Rev. E*, 92(2), 021002 (2015)
- A. K. Dixit, A. Oratis, K. Zinelis, D. Lohse & V. Sanjay, *J. Fluid Mech.*, 1010, A2 (2025) — simulations of viscoelastic Worthington jets and droplets

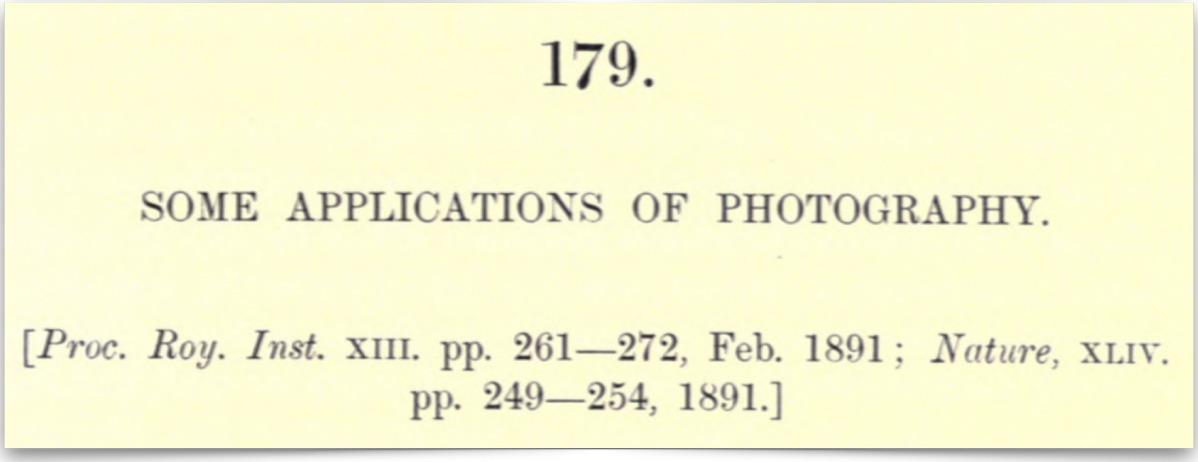
Tiny Additives, Huge Effects

Trace amounts of soft matter (ppm–‰ polymer) can dramatically alter flows and instabilities. This is an engineer's lever in mixing, coating, printing, and food science applications.

A Brief History: Rayleigh (1891) → Taylor–Culick Paradox (1959)

Rayleigh's Pioneering Observations (1891)

179. *SOME APPLICATIONS OF PHOTOGRAPHY.* [*Proc. Roy. Inst.* XIII. pp. 261–272, Feb. 1891; *Nature*, XLIV. pp. 249–254, 1891.]



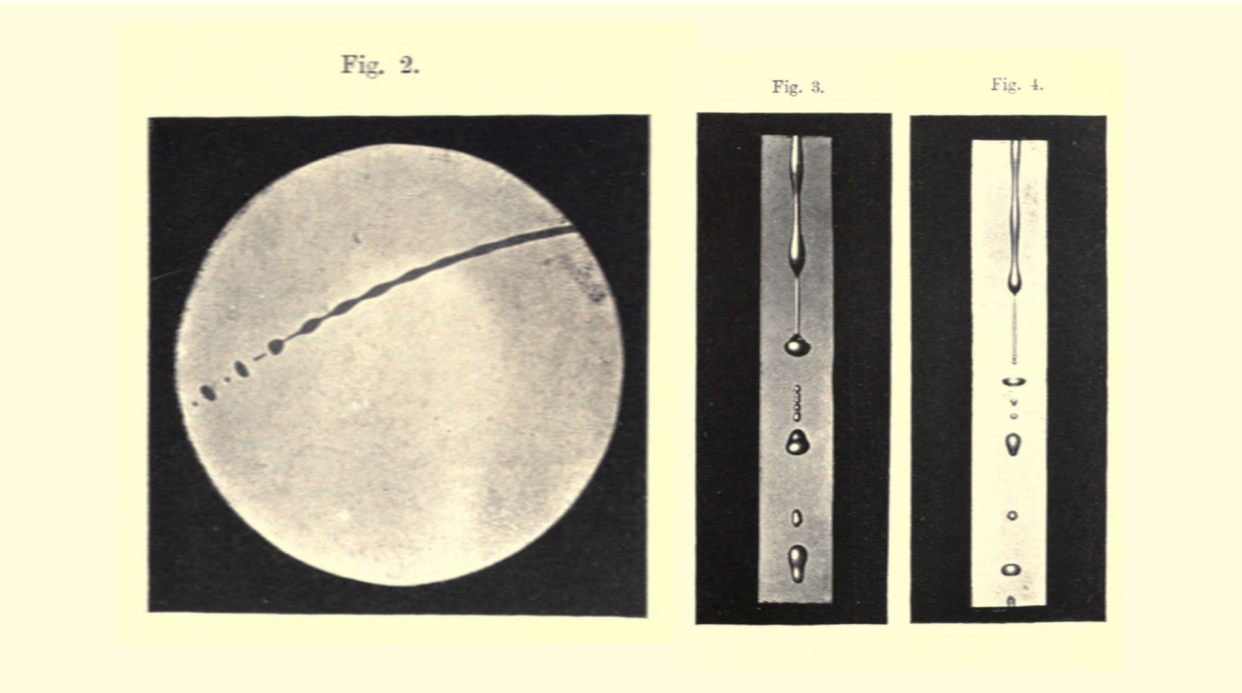
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SOME APPLICATIONS OF PHOTOGRAPHY.

[*Proc. Roy. Inst.* XIII. pp. 261—272, Feb. 1891; *Nature*, XLIV. pp. 249—254, 1891.]

John William Strutt, 3rd Baron Rayleigh, one of the 19th century's greatest physicists, used instantaneous photography to capture:

- Falling jets breaking into droplets (Rayleigh–Plateau instability)

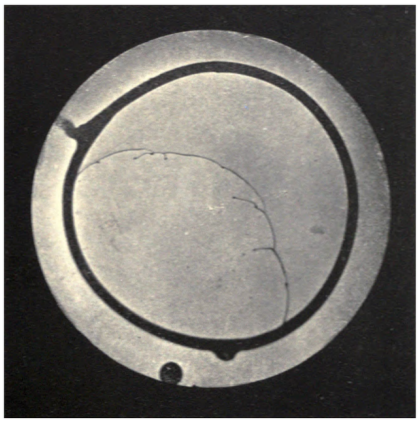


- Liquid stream dynamics and surface tension effects
- Capillary flows driven by interfacial forces

His 1891 paper essentially pioneered the study of liquid dynamics—a cornerstone of soft matter physics.

Taylor–Culick Retraction: The Dupré–Rayleigh Paradox

Figure 7
Taylor–Culick Retractions



When you puncture a soap film, the hole expands rapidly: the film retracts away from the puncture and gathers into a rim at the edge of the growing hole. The film consists of a thin water layer (thickness h_0) with surfactant molecules at both liquid–air interfaces (surface tension γ_{fa} per interface).

The Paradox:

Two seemingly correct approaches give different predictions for retraction speed v_f:

Energy Approach (Dupré; Rayleigh, 1891)

Rate of surface energy decrease = Rate of kinetic energy increase

$$\frac{d}{dt}\left(\frac{1}{2}mv^2\right) = 2\gamma_{fa}(2\pi R)v \tag{9}$$

$$\Rightarrow \quad v_f = \sqrt{\frac{4\gamma_{fa}}{\rho h_0}} \tag{10}$$

Momentum Approach (Taylor, 1959)

Force from surface tension = Rate of momentum change

$$\frac{dP}{dt} = 2\gamma_{fa}(2\pi R) \tag{11}$$

$$\Rightarrow \quad \frac{d}{dt}(mv_f) = 2\gamma_{fa}(2\pi R) \tag{12}$$

$$\Rightarrow \quad v_f = \sqrt{\frac{2\gamma_{fa}}{\rho h_0}} \tag{13}$$

The Factor of 2 Discrepancy

$$\text{Energy method:} \quad v_f = \sqrt{\frac{4\gamma_{fa}}{\rho h_0}} \tag{14}$$

$$\text{Momentum method:} \quad v_f = \sqrt{\frac{2\gamma_{fa}}{\rho h_0}} \tag{15}$$

Question: Which velocity is right?

Think About It

We will return to this question in Lecture 1.5.

Friday Afternoon Experiment: Timescale Matters

Researchers at the University of Naples Federico II studied bubble formation dynamics by varying the air-blowing rate during bubble generation:

Video: `https://www.youtube.com/watch?v=zLucBJkUI5c`

Reference: D. Tammaro, R. Pasquino, M. M. Villone, G. D'Avino, V. Ferraro, E. Di Maio, A. Langella, N. Grizzuti & P. L. Maffettone, *Langmuir*, 34(19), 5646–5654 (2018)

Key Observation:

- Low blowing rate: Bubbles form slowly; film drains gradually; stable bubble structure
- High blowing rate: Bubbles form rapidly; less drainage time; different dynamics and breakup modes

Lesson: In soft matter, protocol and timescale are decisive control parameters. The same material exhibits wildly different behavior depending on how fast you drive it.

Epilogue: Reflection Questions

1. Name one soft-matter phenomenon you now see in everyday life. Why is it soft?
2. What do you want to get out of this course?
3. (*Optional*) Share a 1–2 sentence observation on something from your week that now "looks like soft matter."

Homework (Self-Assessed)

Homework Questions

Q1. Explain, in your own words, why many soft materials have shear moduli orders of magnitude below crystalline solids. (Think about atomic vs mesoscopic length scales and energy sources.)
Q2. Pick an everyday product and identify:

- Its soft-matter classification (foam, gel, polymer solution, colloid, etc.)
- The microstructural ingredient that gives it its desired function

Q3. A polymer coil in a good solvent is stretched and released. Why is the restoring force entropic rather than from stiff bond stretching?
Q4. A thin water stream breaks into drops. Name:

- The instability responsible
- Its thermodynamic driver (what energy is minimized?)
- One engineering application or mitigation strategy (e.g., in printing, mixing, coating)

1.75-Complementary-Questions-Answers

Appendix: Elasticity "Cheat Sheet"

Understanding K, G, E, and ν

Bulk Modulus (K) — Resistance to Volume Change

- Definition: $\Delta V/V = -\Delta p/K$ (uniform compression from all directions)
- Large K means small volume changes under pressure → (nearly) incompressible
- Typical values: Water and rubber both have $K \sim 1$–$2\,\mathrm{GPa}$ → hard to compress

Example: A 1 atm pressure change (100 kPa) on water with $K = 2\,\mathrm{GPa}$ gives $\Delta V/V \approx 5 \times 10^{-5}$ (0.005% volume change)—effectively incompressible in everyday terms.

Shear Modulus (G) — Resistance to Shape Change

- Definition: Stress needed to shear a material at (nearly) constant volume
- Small G → easy to reshape → "soft in shear"
- Typical values: rubbers and gels $G \sim 1\,\mathrm{kPa}$ to $1\,\mathrm{MPa}$; hard matter $G \sim 50$–$200\,\mathrm{GPa}$

Young's Modulus (E) — Uniaxial Stiffness

- Definition: Stress divided by strain in a simple tension/compression test
- Not independent: Combines both shear and bulk responses via the Poisson effect
- Typical values: rubbers and gels $E \approx 3G \sim 3\,\mathrm{kPa}$ to $3\,\mathrm{MPa}$; hard matter $E = 2G(1+\nu) \sim 100$–$500\,\mathrm{GPa}$

Poisson's Ratio (ν) — Lateral Contraction

- Definition: When you stretch a rod, it gets thinner; ν quantifies this lateral contraction
- Incompressible limit: $\nu \to 0.5$ (no volume change under axial load)
- Typical values: Metals $\nu \approx 0.2$–0.35; rubbers $\nu \approx 0.5$

Relations Between Elastic Constants

For isotropic, linear elastic solids, any two constants determine the others:

$$\boxed{E = 2G(1+\nu)} \tag{16}$$

$$\boxed{E = \frac{9KG}{3K+G}} \tag{17}$$

$$\boxed{K = \frac{E}{3(1-2\nu)}} \tag{18}$$

Key Limits:

- Incompressible limit ($\nu \to 0.5$): $K \gg G$ and $E \approx 3G$
  - This is why rubbers and gels can be "soft" (G small) yet "incompressible" (K large)
- Compressible solids (metals): $\nu \approx 0.2$–0.35, so K and G are comparable

Liquids: A Special Case

A Newtonian liquid (water, simple oils) cannot support static shear stress, so:

- $G = 0$ (no static shear modulus)
- K is large (liquids are hard to compress)
- Resistance to shear is purely viscous (η, viscosity), not elastic

The Incompressible Yet Soft (?)

Large K doesn't imply large G.

- Bulk modulus (K): Controls volume resistance under hydrostatic pressure (bonding/packing physics)
- Shear modulus (G): Controls shape resistance at nearly constant volume (entropy in polymer networks; surface tension in foams and emulsions; viscosity in liquids)

Rubbers, gels and liquids are soft in G (easy to reshape), not in K (hard to compress).

- Rubbers/gels: $K \gg G$, $\nu \approx 0.5$, $E \approx 3G$ (for gels, on times too short for the solvent to drain)
- Liquids: K large, $G = 0$ (use η instead), no meaningful E
- Foams: small $G \sim \gamma/R$ from surface tension, and compressible: the gas sets K of order the gas pressure, about 0.1 MPa
- Hard solids: Both K and G large

Quick Reference Table

Property	Rubbers and gels (typical)	Hard Matter (typical)	Meaning
K (Bulk)	1–2 GPa	100–400 GPa	Volume stiffness (hydrostatic)
G (Shear)	1 kPa – 1 MPa	50–200 GPa	Shape stiffness
E (Young's)	3 kPa – 3 MPa ($E \approx 3G$)	100–500 GPa ($E = 2G(1+\nu)$)	Uniaxial stiffness
ν (Poisson)	≈ 0.5	0.2–0.35	Lateral contraction ratio

Each column describes one population of materials, so its entries are mutually consistent: for rubbers and gels, $\nu \approx 0.5$ gives $E \approx 3G$. Foams do not fit this column: their G comes from surface tension and their gas makes them compressible.

Key Takeaways

1. Soft matter is everywhere. From gels and foams to champagne bubbles and mud, soft materials dominate our everyday experience.
2. "Soft" and "incompressible" are not contradictory. Rubbers, gels and other liquid-based soft materials are hard to compress (large K) but easy to reshape (small G). Foams are the exception: their gas makes them compressible.
3. Entropic elasticity is different. In ideal flexible polymer networks, the shear modulus arises from statistical resistance to entropy loss, not stiff bond stretching. At a fixed density n of elastically active strands, $G = nk_BT$ grows in proportion to the absolute temperature; changing the crosslink density changes n. Foams and emulsions are soft for a different reason: their shear modulus comes from surface tension.
4. Time matters. The same soft material behaves differently over different timescales—as a solid at short times, as a fluid at long times. This time-dependent response is central to soft matter physics.
5. Tiny additives, huge effects. Trace polymer additions can dramatically alter interfacial dynamics, instabilities, and flow behavior—an important design lever in engineering and biology.

Further Reading

1. Textbook: Pierre-Gilles de Gennes, *Scaling Concepts in Polymer Physics* (Cornell University Press, 1979) — foundational work on soft matter scaling Link
2. Review: Patrick Oswald & Pierre Pieranski, *Nematic and Cholesteric Liquid Crystals: Concepts and Physical Properties Illustrated by Experiments* (Taylor & Francis, 2005) — detailed liquid crystal physics DOI: 10.1201/9780203023013

3. Modern perspective: Tom McLeish, *Soft Matter: A Very Short Introduction* (Oxford University Press, 2020) — contemporary soft matter overview Link

See the next lecture for a rigorous treatment of the Taylor–Culick paradox and momentum/energy conservation in soft matter flows.

Next Lecture: Lecture 1.5: The Taylor–Culick Paradox

Course Home: Course Overview

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